

REPUBLIC OF THE PHILIPPINES · DEPARTMENT OF EDUCATION

Grade 6 Mathematics

YEAR-END EXAMINATION

SY 2026–2027 · MATATAG

Cumulative · 60 points · Terms 1–3
Includes Table of Specifications and Answer Key for the teacher

Year-End Examination

Name: _____ Section: _____ Date: _____ Score: _____/60

Cumulative examination covering all three terms. Use $\pi = 3.14$ where needed. Show solutions for Part III.

I. Multiple Choice (1 pt each, 30 pts)

1. $14.7 + 8.95 =$
 A. 22.65 B. 23.65 C. 23.55 D. 22.55
2. $6.3 \times 0.4 =$
 A. 2.52 B. 25.2 C. 0.252 D. 2.42
3. $9.6 \div 0.8 =$
 A. 1.2 B. 12 C. 120 D. 0.12
4. $\frac{1}{2} + \frac{1}{3} =$
 A. $\frac{2}{5}$ B. $\frac{5}{6}$ C. $\frac{2}{6}$ D. $\frac{1}{6}$
5. $\frac{2}{3} \times \frac{3}{5} =$
 A. $\frac{2}{5}$ B. $\frac{6}{8}$ C. $\frac{5}{8}$ D. $\frac{2}{8}$
6. $3 \div \frac{1}{2} =$
 A. $1\frac{1}{2}$ B. 6 C. $\frac{3}{2}$ D. 3
7. The GCF of 12 and 18 is
 A. 2 B. 3 C. 6 D. 36
8. The LCM of 4 and 6 is
 A. 12 B. 24 C. 2 D. 10
9. The value of 2^4 is
 A. 8 B. 16 C. 6 D. 32
10. Simplify: $3 + 4 \times 2^2$
 A. 28 B. 19 C. 14 D. 56
11. Which shape tessellates by itself?
 A. circle B. regular hexagon C. regular pentagon D. regular octagon
12. Simplify the ratio 9 : 12.
 A. 3:4 B. 4:3 C. 3:5 D. 9:12
13. Find n: $\frac{3}{5} = \frac{n}{20}$
 A. 4 B. 12 C. 15 D. 9

14. Write 60% as a fraction in lowest terms.

A. $\frac{3}{5}$

B. $\frac{6}{100}$

C. $\frac{2}{3}$

D. $\frac{3}{4}$

15. 30% of 120 =

A. 30

B. 36

C. 40

D. 90

16. A ₱400 item at 25% off costs

A. ₱300

B. ₱100

C. ₱375

D. ₱320

17. Simple interest on ₱2,000 at 5% for 2 years is

A. ₱100

B. ₱200

C. ₱20

D. ₱400

18. An angle of 120° is

A. acute

B. right

C. obtuse

D. straight

19. A mirror image is produced by a

A. translation

B. reflection

C. rotation

D. slide

20. Volume of a $5 \times 4 \times 3$ cm box is

A. 12

B. 60

C. 20

D. 27

21. 1 litre equals

A. 100 mL

B. 1,000 mL

C. 10 mL

D. 10,000 mL

22. A triangle with a 90° angle is a

A. right triangle

B. acute triangle

C. obtuse triangle

D. equilateral triangle

23. Area of a triangle, base 8, height 5, is

A. 40

B. 20

C. 13

D. 26

24. Area of a trapezoid, bases 6 & 10, height 4, is

A. 32

B. 64

C. 16

D. 40

25. The diameter is ____ the radius.

A. half

B. twice

C. equal to

D. thrice

26. Circumference of a circle with $d = 10$ cm is

A. 31.4 cm

B. 15.7 cm

C. 62.8 cm

D. 78.5 cm

27. Area of a circle with $r = 5$ cm is

A. 31.4

B. 78.5

C. 25

D. 15.7

28. The mean of 6, 8, 10 is

A. 8

B. 24

C. 7

D. 9

29. A pie graph shows

A. trends

B. parts of a whole

C. distance

D. speed

30. P(rolling an even number on a die) is

A. $\frac{1}{6}$

B. $\frac{1}{3}$

C. $\frac{1}{2}$

D. 1

II. Computation (2 pts each, 20 pts)

31. $2\frac{3}{4} + 1\frac{1}{2} =$ _____
32. GCF and LCM of 18 and 24 = _____
33. Simplify: $5^2 + (10 - 6) \times 3 =$ _____
34. Divide ₱1,500 in the ratio 1:2:2. = _____
35. 40% of 250 = _____
36. Volume of a rectangular prism $9 \times 4 \times 5$ cm. = _____
37. Area of a parallelogram, base 11 cm, height 6 cm. = _____
38. Circumference of a circle with $r = 7$ cm. = _____
39. Mean, median, and mode of 5, 7, 7, 9, 12. = _____
40. P(red) from a bag of 4 red, 1 blue. = _____

III. Problem Solving (show solutions, 10 pts)

41. (2 pts) Aling Maria bought 2.5 kg of pork at ₱280/kg. How much change from ₱1,000?

42. (2 pts) A ₱1,200 jacket is 30% off. What is its sale price?

43. (3 pts) A circular garden has a radius of 6 m. Find its circumference and its area.

44. (3 pts) Two bells ring every 9 and 12 minutes, starting together at 6:00 a.m. When do they next ring together? Then, of 5 quiz scores 8, 9, 9, 10, 14, find the mean.

Table of Specifications & Answer Key

Year-End Examination · 60 points · Terms 1–3

Table of Specifications

Term / Domain	Items	Pts	%
Term 1 — decimals, fractions, GCF/LCM, exponents, tessellation (NA, MG)	1–11, 31, 32, 33, 41	17	28%
Term 2 — ratio, proportion, percent, angles, transformations, volume (NA, MG)	12–21, 34, 35, 36, 42	18	30%
Term 3 — plane figures, circles, data, probability (MG, DP)	22–30, 37, 38, 39, 40, 43, 44	25	42%
Total	44 items	60	100%

Cognitive spread: Remembering/Understanding $\approx$ 50% (Part I), Applying $\approx$ 33% (Part II), Analyzing/Problem-solving $\approx$ 17% (Part III).

Answer Key — Part I (Multiple Choice)

1	2	3	4	5	6	7	8	9	10
B	A	B	B	A	B	C	A	B	B
11	12	13	14	15	16	17	18	19	20
B	A	B	A	B	A	B	C	B	B
21	22	23	24	25	26	27	28	29	30
B	A	B	A	B	A	B	A	B	C

#10: $3 + 4 \times 4 = 19$ (B). #15: $0.30 \times 120 = 36$ (B). #23: $\frac{1}{2} \times 8 \times 5 = 20$ (B). #24: $\frac{1}{2}(6+10) \times 4 = 32$ (A). #27: $3.14 \times 25 = 78.5$ (B).

Answer Key — Part II (Computation)

31) $4\frac{1}{4}$ 32) GCF = 6, LCM = 72 33) $25 + 12 = 37$ 34) ₱300, ₱600, ₱600 35) 100 36) 180 cm^3 37) 66 cm^2 38) $2 \times 3.14 \times 7 = 43.96 \text{ cm}$ 39) mean = 8, median = 7, mode = 7 40) $\frac{4}{5}$

Answer Key — Part III (Problem Solving)

41) $2.5 \times 280 = \text{₱}700$; change = $1000 - 700 = \text{₱}300$.
 42) discount = $0.30 \times 1200 = \text{₱}360$; sale price = $1200 - 360 = \text{₱}840$.
 43) C = $2 \times 3.14 \times 6 = \mathbf{37.68 \text{ m}}$; A = $3.14 \times 6 \times 6 = \mathbf{113.04 \text{ m}^2}$.
 44) LCM(9,12) = 36 min $\rightarrow$ **6:36 a.m.**; mean of 8,9,9,10,14 = $50 \div 5 = \mathbf{10}$.